Coursework CST2550 reset

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Project report

1. Introduction

This project goal to manage everything using C# and .NET to help small shops manage activities more efficiently. This allows users to manage products, suppliers, stock records and sales using a SQL Server database with Entity Framework Core. A custom Binary Search Tree implemented to improve product searching by barcode. This allows products to be searched quickly while also demonstrating the use of a custom data structure instead of depend on built in collection classes.

The system includes features like product management, supplier management, stock updates, sales processing, low-stock alerts and reporting. Validation has also been added to prevent invalid data like duplicate barcodes empty fields and negative prices. Overall the project provides a simple supermarket management for shops and retails.

2. Data Structures and Algorithms

I used a custom Binary Search Tree to store products in memory. I choose this data structure because it makes searching for products by barcode faster. The project also uses SQL Server with Entity Framework Core to store data permanently. The Binary Search Tree is used for searching while the database stores all product supplier and sales information. This project supports two search methods as well where products can be searched by barcode or by product name.

3. Time Complexity Analysis

The Binary Search Tree improves the speed of searching products by barcode. However, some operations are faster than others depending on how the tree is organised. The table shows the time complexity of the main operations used in this project.

Operation	Time Complexity	Explanation
Insert product	O(log n) average	The product is inserted into the correct position in the Binary Search Tree

Search by barcode	O(log n) average	The search follows one path in the tree until the product is found
Search by name	O(n)	Every product must be checked because the tree is organised by barcode, not by name
Update product	O(log n) average	The product is found by barcode first and then its details are updated
Delete Product	O(log n) average	The product is searched first before it is removed from the tree
Low Stock Report	O(n)	Every product is checked to see if its stock is below the required level

Pseudo Code

Insert Product:

```
START
Input product
If tree is empty
   Create root node
Else
   Compare barcode
   Move left or right
   Insert product into correct position
End If
STOP
```

Search by Barcode

```
START
Input barcode
```

Begin at root

While current node exists

If barcode matches

Return product

Else If barcode is smaller

Move to left child

Else

Move to right child

End While

Return product not found

STOP

Generate Low Stock Report:

START

Visit every product

If stock is below threshold

Display product

Repeat until all products have been checked

STOP

4. Testing

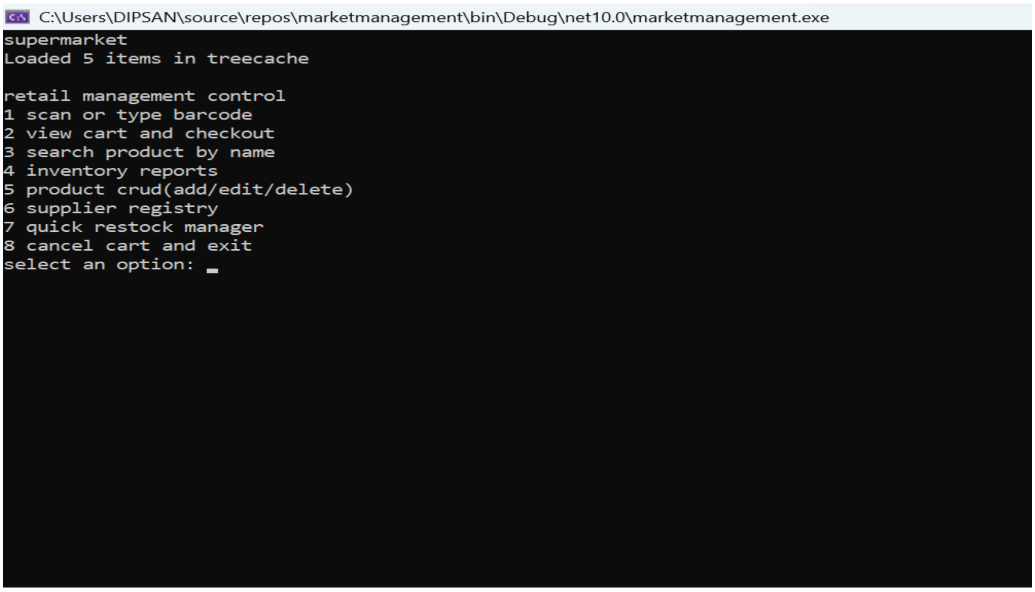
The application is tested using both automated unit testing and manual testing. MStest was used to test the main functions of the Binary Search Tree, while the remaining features were tested by running the application and checking that each function worked correctly. All of these tests passed successfully.

The remaining application features, including validation and supermarket workflows, were tested manually by using the console application. This included checking duplicate barcode validation, invalid prices, negative stock values, supplier registration, sales processing and low stock alerts.

Test Case	Testing Method	Result
Add Product	MSTest	Passed
Search product by barcode	MSTest	Passed
Update product	MSTest	Passed
Search Unknown product	MSTest	Passed
Delete Product	MSTest	Passed
Duplicate barcode	Manual Testing	Passed
Invalid price	Manual Testing	Passed
Negative stock	Manual Testing	Passed
Supplier registry	Manual Testing	Passed
Sales transaction	Manual Testing	Passed
Low stock alert	Manual Testing	Passed

4.1Testing screenshot

- Main menu

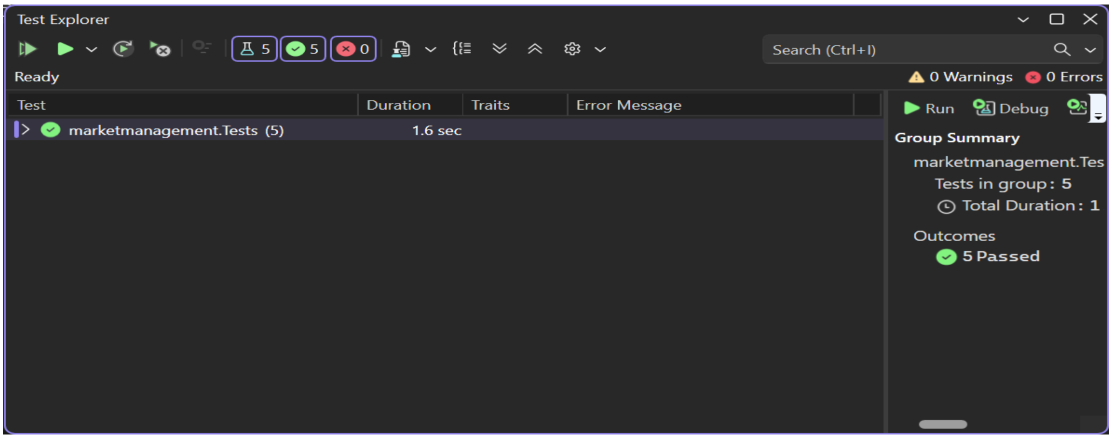


- Search product by name

```
retail management control
1 scan or type barcode
2 view cart and checkout
3 search product by name
4 inventory reports
5 product crud(add/edit/delete)
6 supplier registry
7 quick restock manager
8 cancel cart and exit
select an option: 3
enter item name to searchmilk

tree traversal search results for'milk'
match found Barcode50123456 | Name: milk 21 | Price: £2.50 | Stock: 45
```

- MSTest results



- **Inventory report**

```
retail management control
1 scan or type barcode
2 view cart and checkout
3 search product by name
4 inventory reports
5 product crud(add/edit/delete)
6 supplier registry
7 quick restock manager
8 cancel cart and exit
select an option: 4

low stock alert log (threshold: < 50 units)
ALERT: milk 21 (Barcode: 50123456) is running low! Only 45 units left.

 expiration & restock schedule
Product: bread | Barcode: 111111 | Scheduled Date: 2026-07-06
Product: ice cream | Barcode: 222222 | Scheduled Date: 2026-07-06
Product: ice | Barcode: 333333 | Scheduled Date: 2026-07-09
Product: milk 21 | Barcode: 50123456 | Scheduled Date: 2026-06-29
Product: apple | Barcode: 777777 | Scheduled Date: 2026-07-10

 sales log
Sale ID: 1 | Date: 2026-06-25 12:00 | Total Revenue: £0.50
Sale ID: 2 | Date: 2026-06-25 12:31 | Total Revenue: £70.00
Sale ID: 3 | Date: 2026-06-26 13:18 | Total Revenue: £4.35
Sale ID: 4 | Date: 2026-06-29 01:43 | Total Revenue: £14.50
```

5. Conclusion

This project helped me understand how Binary Search Trees, SQL Server and Entity Framework Core can be used together in one application. The supermarket system can manage products, suppliers, stock and sales while also providing searching and reporting features. Overall, the project met its main objectives and worked as expected. If I had more time, I would improve the user interface and add features such as barcode scanner support and receipt printing.

6. References

Microsoft (2025) C# documentation. Available at: https://learn.microsoft.com/dotnet/csharp/ (Accessed: 1 July 2026).

Microsoft (2025) Entity Framework Core documentation. Available at: https://learn.microsoft.com/ef/core/ (Accessed: 1 July 2026).

Microsoft (2025) SQL Server documentation. Available at: https://learn.microsoft.com/sql/ (Accessed: 1 July 2026).

Microsoft (2025) MSTest unit testing documentation. Available at: https://learn.microsoft.com/dotnet/core/testing/unit-testing-with-mstest (Accessed: 1 July 2026).